

DATA PIPELINE_02: BLOB to SQLDB (ADF USING POWERQUERY)

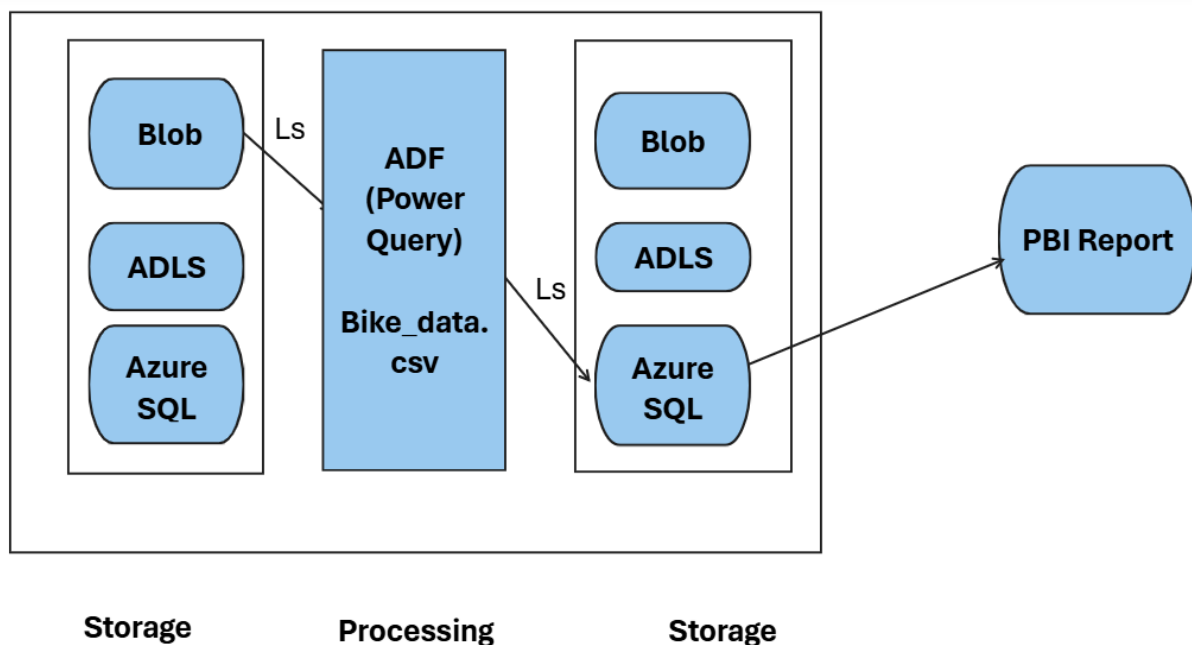
Project Objective:

To extract raw data which is stored in Azure Blob Storage, transform it using power query in Azure Data Factory, and load it into Azure SQL Database.

Process:

Copy File from Blob Storage → Transform (ADF using powerquery) → Load into SQL Database (via ADF)

Architecture:



What Is BLOB:

Azure Blob Storage is Microsoft's cloud-based object storage solution designed to store massive amounts of unstructured data. It is ideal for files that don't fit into traditional databases, such as images, videos, audio, backups, and log files.

1)TO CREATE BLOB STORAGE:

Step 1: Sign in to Azure Portal ----Go to <https://portal.azure.com> and Sign in with your Microsoft/Azure account

Step 2: 3. Create an Azure Storage Account ---Blob storage called a storage account — like a big folder in the cloud.

Azure services



- Click "Create a resource"

Home

Create a resource

I need a new low-cost VM

- Search for "Storage account" and click Create

Fill basic details:

Subscription: Choose your subscription: Azure Subscription1

Resource Group: Select or create a new one : nk-rg

Storage account name: Unique name (lowercase only) : nkblobsa

Region: Pick a location close to your users

Primary service : Azure Blob or Azure Data Lake Storage Gen2

Performance: Standard (default is fine)

Redundancy: Choose Locally-redundant storage (LRS) for basic use.

Fill Advanced details:

Access Tier: Hot, Cool, or Cold based on usage.

Fill Networking Details:

Network Access: Enable Public access from all networks (for practice).

Routing Preference: microsoft Network Routing.

Fill Data Protection details:

Soft delete for blobs: Enable (recommended)

Soft delete for containers: Optional.

Soft delete for file shares: Optional

Click Review + Create, then Create

Home > Create a resource

Create a storage account

Basics Advanced Networking Data protection Security Encryption Tags **Review + create**

[View automation template](#)

Basics

Subscription	NK
Resource group	NK-RG
Location	South India
Storage account name	nkblobsa
Primary service	Azure Blob Storage or Azure Data Lake Storage
Performance	Standard
Replication	Read-access geo-redundant storage (RA-GRS)

Advanced

Enable hierarchical namespace	Disabled
Enable SFTP	Disabled
Enable network file system v3	Disabled

[Previous](#) [Next](#) [Create](#)

*** Initializing deployment...
Initializing template deployment to resource group.

Sucessfully created blob storage

Home

nkblobsa_1782196969836 | Overview

Deployment

Search Delete Cancel Redeploy Download Refresh

Overview

- Inputs
- Outputs
- Template

✓ Your deployment is complete

Deployment name: nkblobsa_1782196969836
Subscription: [NK](#)
Resource group: [NK-RG](#)

Start time: 23/6/2026, 12:14:19 pm
Correlation ID: e2223130-d522-4815-9f82-6354d4cff0d3

Deployment details

Next steps

[Go to resource](#)

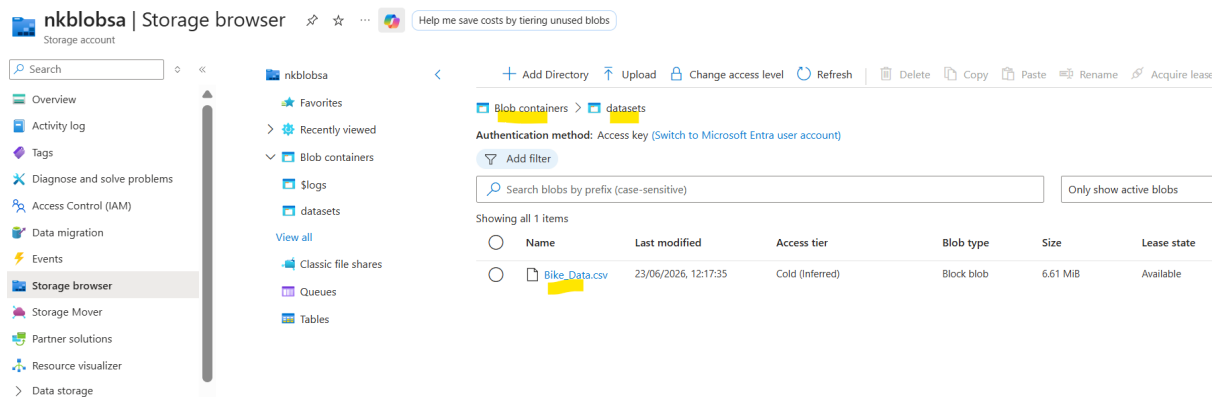
Give feedback

[Tell us about your experience with deployment](#)

Create a Blob Container:

Containers are like folders inside the storage account to organize files.

- Click "+ Container"
- Give it a name: datasets
- Click Create
- Upload the raw data



2) TO CREATE AZURE SQL DB:

Create a SQL Database:

- Click "Create a resource" (top left)
- Search for "SQL Database", then click Create
- Fill in the basics:
 - o Subscription: Choose your subscription

Resource Group: Select the one you created

Database name: Give database name

Server: Click "Create new"

Project details

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription * ⓘ NK ▼

Resource group * ⓘ NK-RG ▼

[Create new](#)

Create a SQL Server

- This is where your database will present.
- Click Create new (under SQL Server)

Server name: (must be unique)

Location: Same as resource group

Admin login: Create a username

Password: Set a strong password o Click OK to return to the SQL DB creation form.

Configure Database:

Choose Basic pricing (DTU Basic)

Database details

Enter required settings for this database, including picking a logical server and configuring the compute and storage resources

Database name *

Server * ⓘ 

[Create new](#)

Enable Public Access (Firewall Rule):

Connectivity Method: Public Endpoint

In the Networking tab, under Firewall rules, select: Allow Azure services and resources to access this server: yes

Network connectivity

Choose an option for configuring connectivity to your server via public endpoint or private endpoint. Choosing no access creates with defaults and you can configure connection method after server creation. [Learn more](#) 

Connectivity method * ⓘ

☐ No access

☒ Public endpoint

☐ Private endpoint

Firewall rules

Setting 'Allow Azure services and resources to access this server' to Yes allows communications from all resources inside the Azure boundary, that may or may not be part of your subscription. [Learn more](#) 

Setting 'Add current client IP address' to Yes will add an entry for your client IP address to the server firewall.

Allow Azure services and resources to access this server *

Add current client IP address *

- Click Review + create
- Wait for validation
- Click Create

Home

Microsoft.SQLDatabase.newDatabaseNewServer_d5bde4d31f7d402d86326 | Overview

Search

Overview

Inputs

Outputs

Template

Deployment is in progress

Deployment name : Microsoft.SQLDatabase.newDatabaseNewServer_d5bd... Start time : 23/6/2026, 12:38:35 pm
Subscription : NK Correlation ID : 2ed2ff7e-3ef5-4722-b0c5-8e2d61ef9170
Resource group : NK-RG

Deployment details

Resource	Type	Status	Operation details
nkserver	Microsoft.Sql/servers	Accepted	Operation details

Sucessfully created the sql database

Home

Microsoft.SQLDatabase.newDatabaseNewServer_d5bde4d31f7d402d86326 | Overview

Search

Overview

Inputs

Outputs

Template

Your deployment is complete

Deployment name : Microsoft.SQLDatabase.newDatabaseNewServer_d5bd... Start time : 23/6/2026, 12:38:45 pm
Subscription : NK Correlation ID : 2ed2ff7e-3ef5-4722-b0c5-8e2d61ef9170
Resource group : NK-RG

Deployment details

Next steps

[Go to resource](#)

Go to Resource—Query editor(preview)---Provide the Sql authentication details(username and password)

Home > Microsoft.SQLDatabase.newDatabaseNewServer_d5bde4d31f7d402d86326 | Overview > free-sql-db-4012917 (nkserver/free-sql-db-4012917)

free-sql-db-4012917 (nkserver/free-sql-db-4012917) | Query editor (preview)

Search

Feedback

Classic experience

Overview

Activity log

Tags

Diagnose and solve problems

Query editor (preview)

Mirror database in Fabric (preview)

Resource visualizer

Settings

Data management

Integrations

Power Platform

Security

Intelligent performance

Monitoring

Add or remove favorites by pressing Ctrl+Shift+F

Run queries on your database directly in the browser [Learn more](#)

Connecting to free-sql-db-4012917 on nkserver

Microsoft Entra authentication

SQL authentication

Username *

Enter username

Password *

Enter password

Connect

[Forgot password? Reset in logical server settings](#)

 **free-sql-db-4012917 (nkserver/free-sql-db-4012917)** | Query editor (preview) ☆ ...
SQL database

◇ << <

 New query  Templates ▾  Open in ▾  Feedback  Sign out  Classic experience

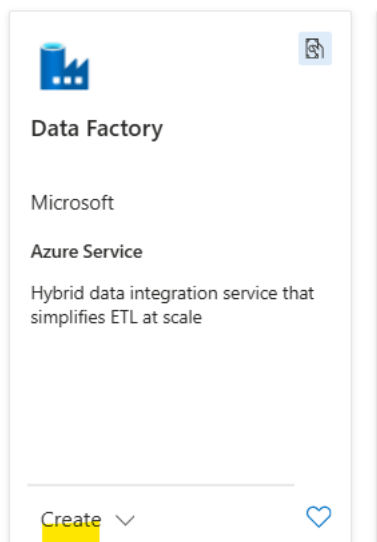
 Overview
 Activity log
 Tags
 Diagnose and solve problems
 **Query editor (preview)**
 Mirror database in Fabric (preview)
 Resource visualizer
> Settings
> Data management
> Integrations

Explorer <<
 ≡
▽ free-sql-db-4012917
 >  dbo
 ▽  Queries


Start exploring
Write and run SQL queries, create database objects
preview your

Create Azure Data Factory :

- Search for Data Factory > Click Create



Fill in the basic details required

Basics

Subscription

Resource group: NK-RG

Name: nk-azuredataf

Region: select region

Version: V2

Networking:

Connect via:Public endpoint

Project details

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription * ⓘ

NK

Resource group * ⓘ

NK-RG

Create new

Instance details

Name * ⓘ

nk-azuredatf

Region * ⓘ

(Asia Pacific) South India

Version * ⓘ

V2

Home > Create a resource > Marketplace

Create Data Factory ...

- Basics
- Git configuration
- Networking
- Advanced
- Tags
- Review + create

 [View automation template](#)

Basics

Subscription	NK
Resource group	NK-RG
Name	nk-azuredatf
Region	South India
Version	V2

Networking

Connect via	Public endpoint
-------------	-----------------

Home



Microsoft.DataFactory-20260623124515 | Overview ...

Deployment

Search



Delete



Cancel



Redeploy



Download



Refresh

Overview

Inputs

Outputs

Template



Your deployment is complete



Deployment name : Microsoft.DataFactory-20260623124515

Subscription : NK

Resource group : NK-RG

Start time : 23/6/202

Correlation ID : 364d24b



Deployment details



Next steps

[Go to resource](#)

Successfully created azure data factory

Resources

Recommendations



Filter for any field...

Type equals all

Location equals all

+ Add filter



Name ↑

Type

Location



free-sql-db-4012917 (nkserver/free-sql-db-4012917)



SQL database

South India



nk-azuredat



Data factory (V2)

South India



nkblobsa



Storage account

South India



nkserver



SQL server

South India



Delete

Resource group (move) : [NK-RG](#)

Type : Data factory (V2)

Status : Succeeded

Getting started : [Quick start](#)

Location : South India

Subscription (move) : [NK](#)

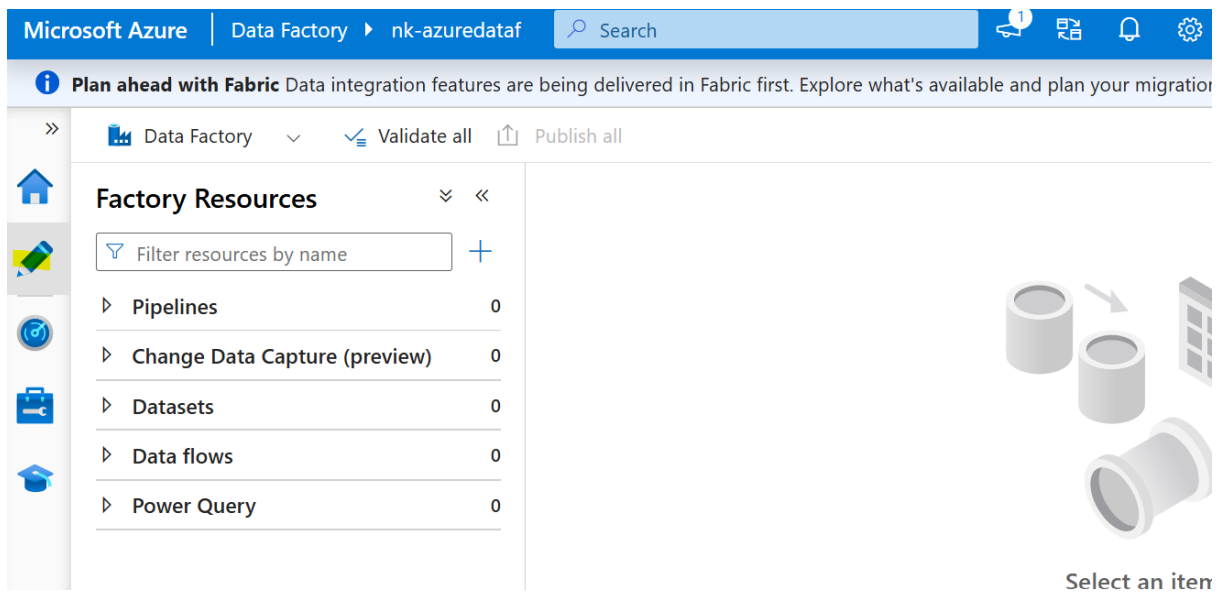
Subscription ID : 5bd8f1cd-03ad-48fa-8410-6e58274fc11f



Azure Data Factory Studio

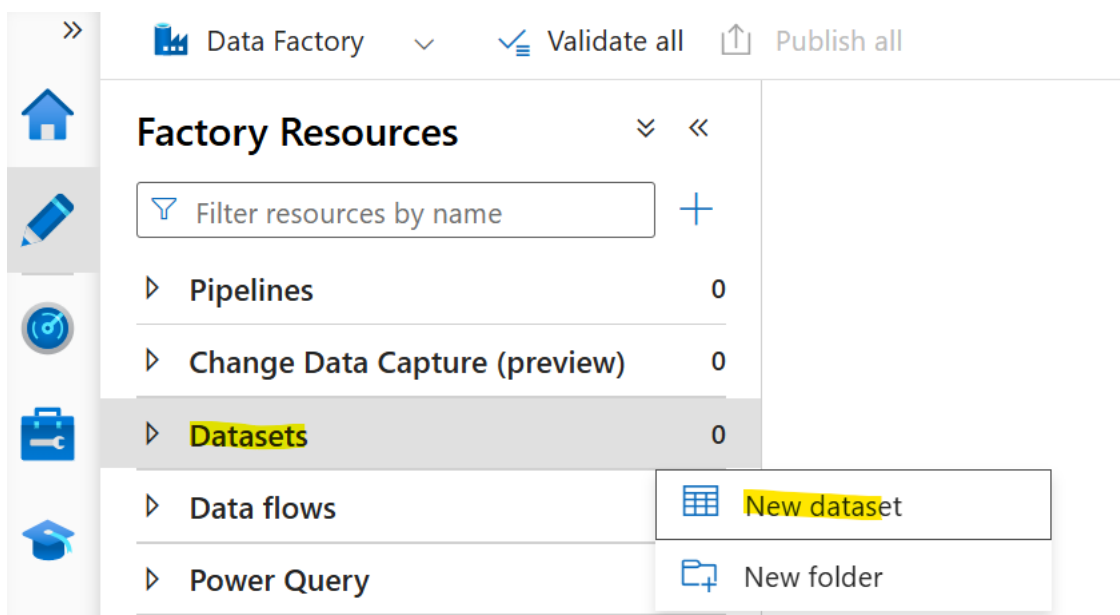
[Launch studio](#)

ADF Studio:



Step 1 in ADF:

We need to create a copy of the raw data from the storage account



Click on the new data set and select from which storage are we taking the copy of the raw data

New dataset

In pipeline activities and data flows, reference a dataset to specify the location and structure of your data within a data store. [Learn more](#) 

Select a data store

 Search

<

All

Azure

Database

File

Generic protocol

>


Azure AI Search


Azure Blob Storage


Azure Cosmos DB for
MongoDB

Continue

Cancel

Select the format of the data and click on continue

Select format

Choose the format type of your data

 Avro	 Binary	 DelimitedText
	 JSON	

Continue

Back

Cancel

A new popup displays give the name and add the linked service to push the copy data from storage

Set properties

Name

InputData

Linked service *

Select...

First row as header



OK

Back

Cancel

Create the linked service :

1. Go to your ADF → Click Launch Studio 2. In left menu → Click Manage (gear icon)
3. Under Linked Services → Click + New
2. For Azure Blob Storage • Choose Azure Blob Storage • Auth type: Account key or Azure AD • Select your storage account • Click Create

Set properties

Name

InputData

Linked service *

blob2adf

File path

datasets / Directory / Bike_Data.csv

First row as header

☒

Import schema

☒ From connection/store ☐ From sample file ☐ None

OK

Back

Cancel

Plan ahead with Fabric Data integration features are

>>

Data Factory

Validate all

Factory Resources

Filter resources by name

Pipelines0

Change Data Capture (preview)0

Datasets1

InputData

Data flows0

Power Query0

Click on publish all to save the changes in adf

Plan ahead with Fabric Data integration features are being delivered in Fabric first. Explore what's available and plan your migration to Fabric. [Start exploring](#)

Data Factory Validate all **Publish all** 1

Factory Resources

Filter resources by name

- Pipelines 0
- Change Data Capture (preview) 0
- Datasets 1**
 - InputData**
- Data flows 0
- Power Query 0

InputData

DelimitedText
InputData

CSV

Connection Schema Parameters

Linked service * blob2adf

Test connection Edit + New [Learn more](#)

File path * datasets / Directory

Compression type No compression

Column delimiter ① Comma (,)

Row delimiter ① Default (\r\n, or \r\n)

Encoding ① Default(UTF-8)

We can also preview the data

Connection Schema Parameters

blob2adf

ew [Learn more](#)

tasets / Directory / Bike_Data.csv Browse **Preview data**

compression

mma (,)

efault (\r\n, or \r\n)

efault(UTF-8)

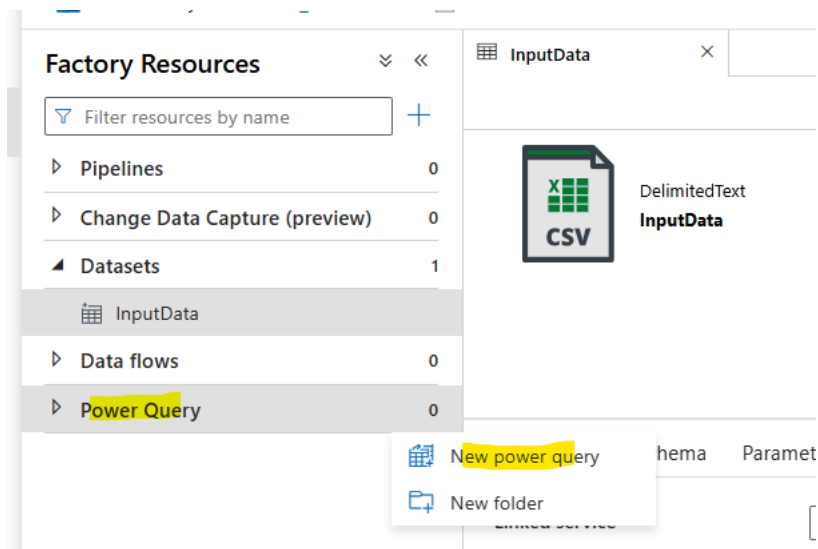
Preview data

Linked service: blob2adf

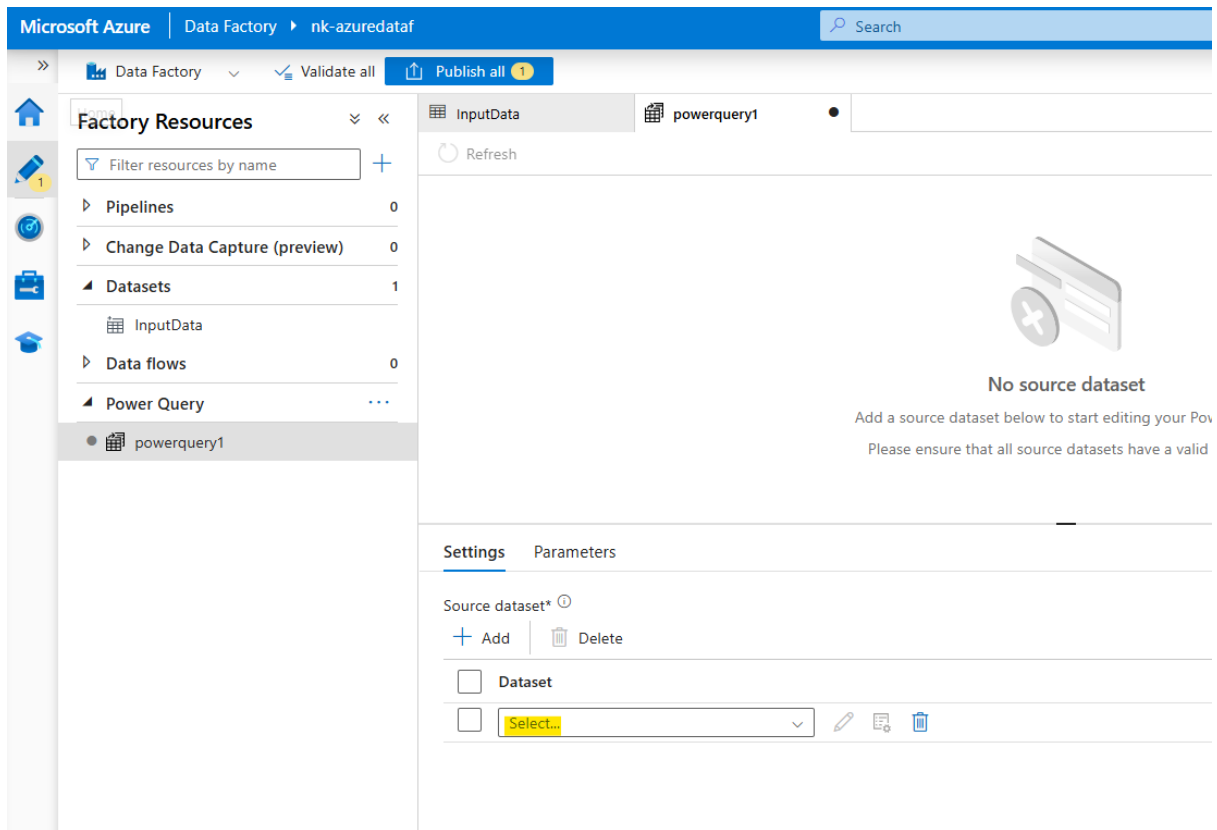
Object: Bike_Data.csv

	Region	Country	Customer	BusinessSegment	Category	Model	Color	SalesDate	ListPrice
1	North America	United States	Professional Containers and Packaging Co.	Clothing	Tights	Women's Tights	Black	22-08-2018	74.99
2	North America	United States	Professional Containers and Packaging Co.	Clothing	Gloves	Full-Finger Gloves	Black	17-08-2018	37.99
3	North America	United States	Professional Containers and Packaging Co.	Bikes	Mountain Bikes	Mountain-200	Black	14-08-2018	2294.9
4	North	United	Professional Containers and	Clothing	Jerseys	Long-Sleeve	Multi	13-08-	49.99

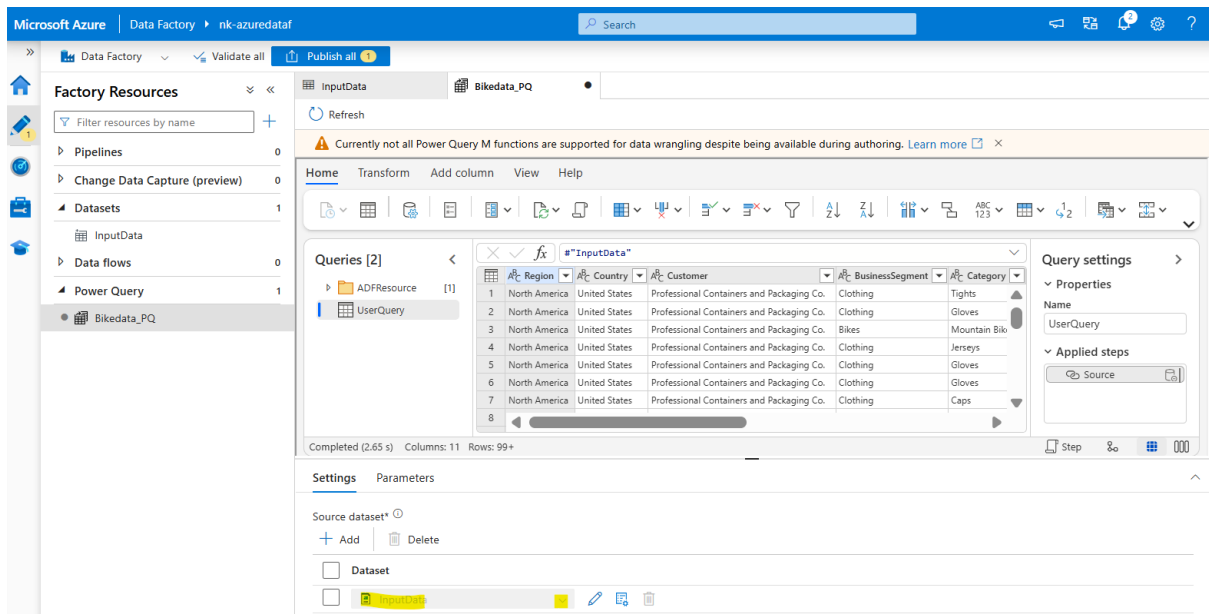
Create power query for Transformations:



Now we can select the data which we copied from blob storage to make transformations in powerquery



- Give the data ---setting---select—dropdown
- Once we select the input data it automatically previews the data



To change the datatypes in powerquery---click the column---left side we can see ABC ---click to change the datatype as required

	Model	Color	SalesDate	ListPrice	UnitPrice	OrderQty
1	Men's Tights	Black	22-08-2018	74		
2	finger Gloves	Black	17-08-2018	37		
3	tain-200	Black	14-08-2018	22		
4	Sleeve Logo Jersey	Multi	13-08-2018	49		
5	finger Gloves	Black	09-08-2018	37		
6	finger Gloves	Black	06-08-2018	37		
7	g Cap	Multi	06-08-2018	8.5		
8	finger Gloves	Black	24-05-2018	37		
9	finger Gloves	Black	11-05-2018	37		
10	tain-200	Black	08-05-2018	22		
11	Men's Tights	Black	01-03-2018	74		
12	g Cap	Multi	23-02-2018	8.5		
13	finger Gloves	Black	22-02-2018	37		
14	Sleeve Logo Jersey	Multi	19-02-2018	49		
15	finger Gloves	Black	13-02-2018	37		
16	finger Gloves	Black	11-02-2018	37.99	22.79	9

1)Adding new columns (cost):

select the unitprice,orderqty in add column and multiply---rename the column name as cost

2)Adding new columns (Sales):

select the Listprice,orderqty in add column and multiply---rename the column name as Sales

3)Adding new columns (Profit):

select the Sales,Cost in add column and Subtract---rename the column name as Profit

All the 3 new columns

Home Transform **Add column** View Help

Queries [2] <

- ADFRsource [1]
- UserQuery

Table.RenameColumns("#Inserted subtraction", {"Subtraction", "Profit"})

	SalesDate	1.2 ListPrice	1.2 UnitPrice	1.2 OrderQty	1.2 Cost	1.2 Sales	1.2 Profit
1	22-08-2018	74.99	44.99	1	44.99	74.99	30
2	17-08-2018	37.99	22.79	2	45.58	75.98	30.4
3	14-08-2018	2294.99	1229.46	1	1229.46	2294.99	1065.53
4	13-08-2018	49.99	28.84	6	173.04	299.94	126.9
5	09-08-2018	37.99	22.79	4	91.16	151.96	60.8
6	06-08-2018	37.99	22.79	6	136.74	227.94	91.2
7	06-08-2018	8.99	5.19	2	10.38	17.98	7.6
8	24-05-2018	37.99	22.79	1	22.79	37.99	15.2
9	11-05-2018	37.99	22.79	5	113.95	189.95	76
10	08-05-2018	2294.99	1229.46	2	2458.92	4589.98	2131.06
11	01-03-2018	74.99	44.99	4	179.96	299.96	120
12	23-02-2018	8.99	5.19	4	20.76	35.96	15.2
13	22-02-2018	37.99	22.79	1	22.79	37.99	15.2
14	19-02-2018	49.99	28.84	4	115.36	199.96	84.6
15	13-02-2018	37.99	22.79	2	45.58	75.98	30.4
16	11-02-2018	37.99	22.79	9	205.11	341.91	136.8
17							

We can remove the unnecessary columns by selecting the columns and click remove

Home Transform **Add column** View Help

Queries [2] <

- ADFRsource [1]
- UserQuery

Table.RenameColumns("#Inserted subtraction", {"Subtraction", "Profit"})

	SalesDate	1.2 ListP	1.2 Cost	1.2 Sales	1.2 Profit
1	22-08-2018		44.99	74.99	30
2	17-08-2018		45.58	75.98	30.4
3	14-08-2018		1229.46	2294.99	1065.53
4	13-08-2018		173.04	299.94	126.9
5	09-08-2018		91.16	151.96	60.8
6	06-08-2018		136.74	227.94	91.2
7	06-08-2018		10.38	17.98	7.6
8	24-05-2018		22.79	37.99	15.2
9	11-05-2018		113.95	189.95	76
10	08-05-2018		2458.92	4589.98	2131.06
11	01-03-2018		179.96	299.96	120
12	23-02-2018		20.76	35.96	15.2
13	22-02-2018		22.79	37.99	15.2
14	19-02-2018		115.36	199.96	84.6
15	13-02-2018		45.58	75.98	30.4
16	11-02-2018		205.11	341.91	136.8
17					

Columns: 14 Rows: 99+

Query settings >

- Properties
 - Name: UserQuery
- Applied steps
 - Source
 - Changed c...
 - Inserted m...
 - Renamed c...
 - Inserted m...
 - Renamed c...
 - Inserted su...
 - Renamed c...

Now I wanted to replace the values in the country column:

Select column---right click---replace the values---basic---give the value name---replace with the new value

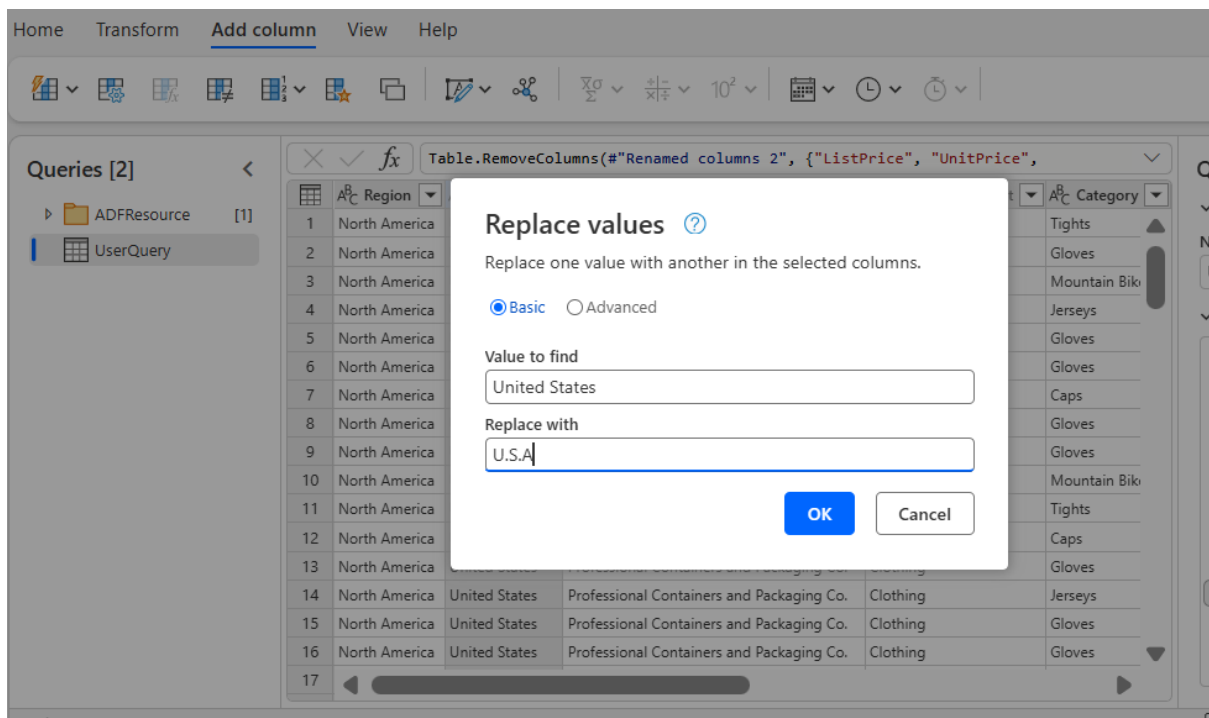
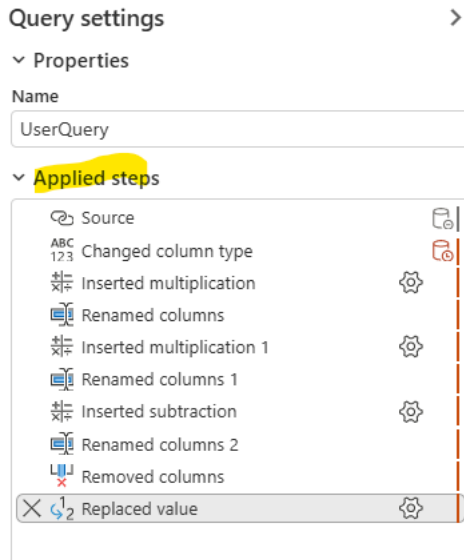


Table.ReplaceValue("#Removed columns", "United States", "U.S.A", Replacer.

	Region	Country	Customer	BusinessSegment	Category
1	North America	U.S.A	Professional Containers and Packaging Co.	Clothing	Tights
2	North America	U.S.A	Professional Containers and Packaging Co.	Clothing	Gloves
3	North America	U.S.A	Professional Containers and Packaging Co.	Bikes	Mountain Biko
4	North America	U.S.A	Professional Containers and Packaging Co.	Clothing	Jerseys
5	North America	U.S.A	Professional Containers and Packaging Co.	Clothing	Gloves
6	North America	U.S.A	Professional Containers and Packaging Co.	Clothing	Gloves
7	North America	U.S.A	Professional Containers and Packaging Co.	Clothing	Caps
8	North America	U.S.A	Professional Containers and Packaging Co.	Clothing	Gloves
9	North America	U.S.A	Professional Containers and Packaging Co.	Clothing	Gloves
10	North America	U.S.A	Professional Containers and Packaging Co.	Bikes	Mountain Biko
11	North America	U.S.A	Professional Containers and Packaging Co.	Clothing	Tights
12	North America	U.S.A	Professional Containers and Packaging Co.	Clothing	Caps
13	North America	U.S.A	Professional Containers and Packaging Co.	Clothing	Gloves
14	North America	U.S.A	Professional Containers and Packaging Co.	Clothing	Jerseys
15	North America	U.S.A	Professional Containers and Packaging Co.	Clothing	Gloves
16	North America	U.S.A	Professional Containers and Packaging Co.	Clothing	Gloves
17					

Note:

Whatever the changes that we are making in powerquery everything is mentioned in the right side of the menu---appliedsteps



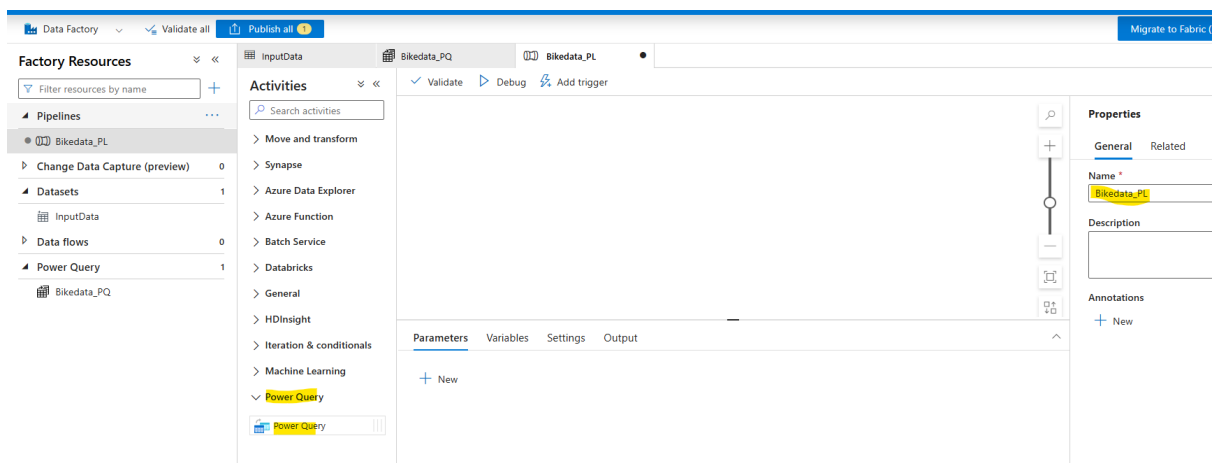
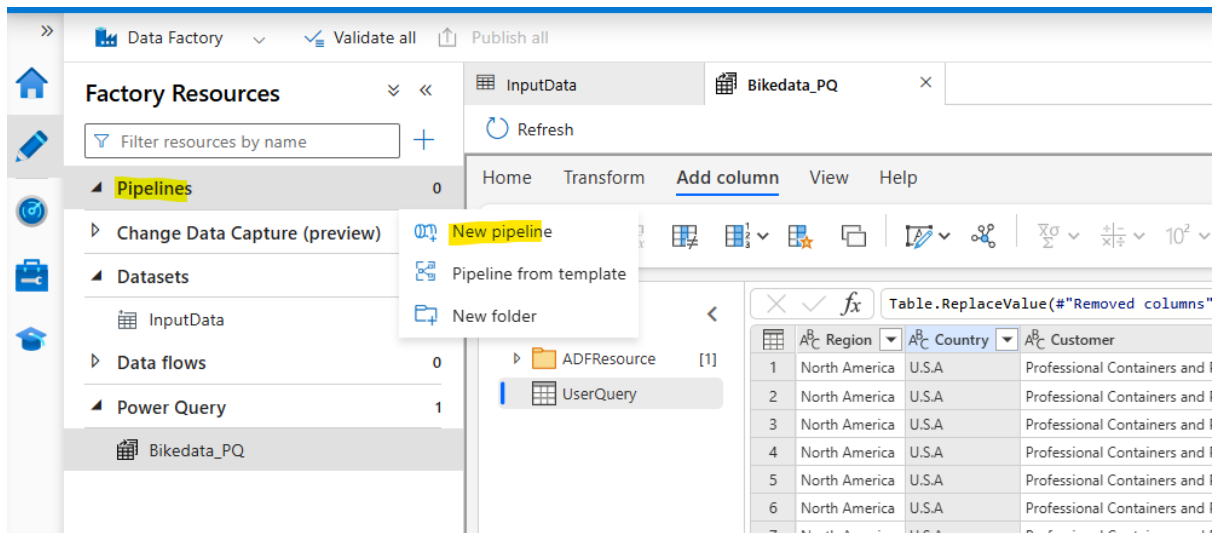
*****Make sure the changes are published after the transformations

	Region	Country	Customer	Clot
1	North America	U.S.A	Professional Containers and Packaging Co.	Clot
2	North America	U.S.A	Professional Containers and Packaging Co.	Clot
3	North America	U.S.A	Professional Containers and Packaging Co.	Bike
4	North America	U.S.A	Professional Containers and Packaging Co.	Clot
5	North America	U.S.A	Professional Containers and Packaging Co.	Clot
6	North America	U.S.A	Professional Containers and Packaging Co.	Clot
7	North America	U.S.A	Professional Containers and Packaging Co.	Clot
8	North America	U.S.A	Professional Containers and Packaging Co.	Clot
9	North America	U.S.A	Professional Containers and Packaging Co.	Clot
10	North America	U.S.A	Professional Containers and Packaging Co.	Bike
11	North America	U.S.A	Professional Containers and Packaging Co.	Clot
12	North America	U.S.A	Professional Containers and Packaging Co.	Clot
13	North America	U.S.A	Professional Containers and Packaging Co.	Clot
14	North America	U.S.A	Professional Containers and Packaging Co.	Clot

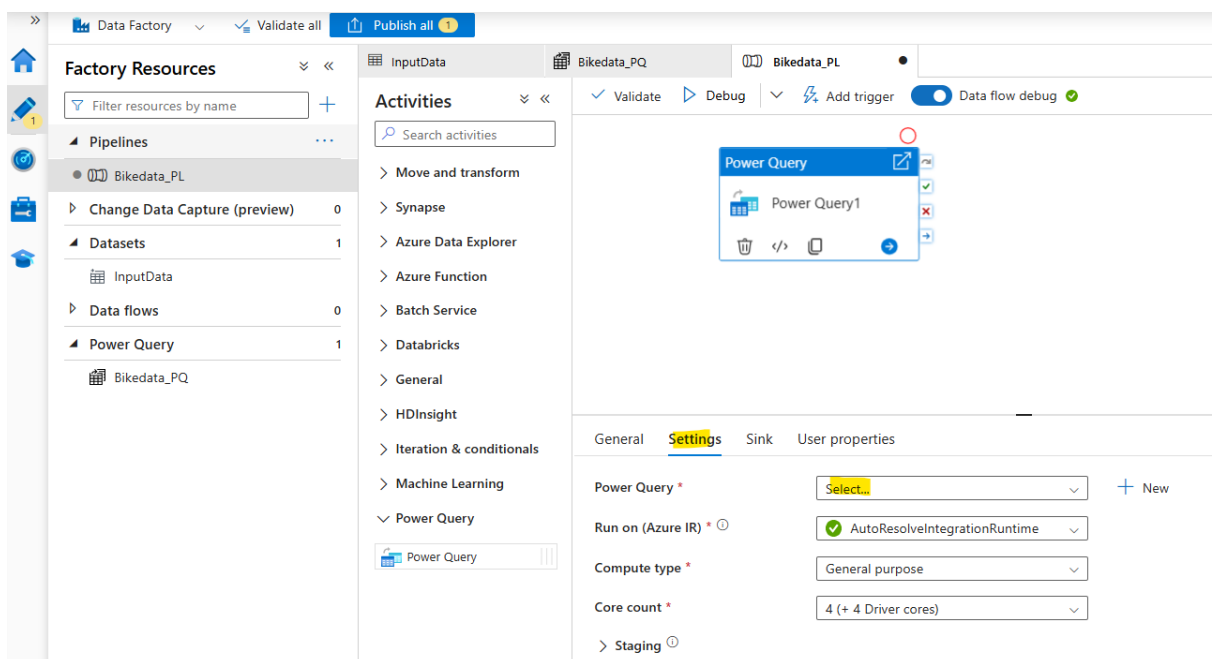
***Note:

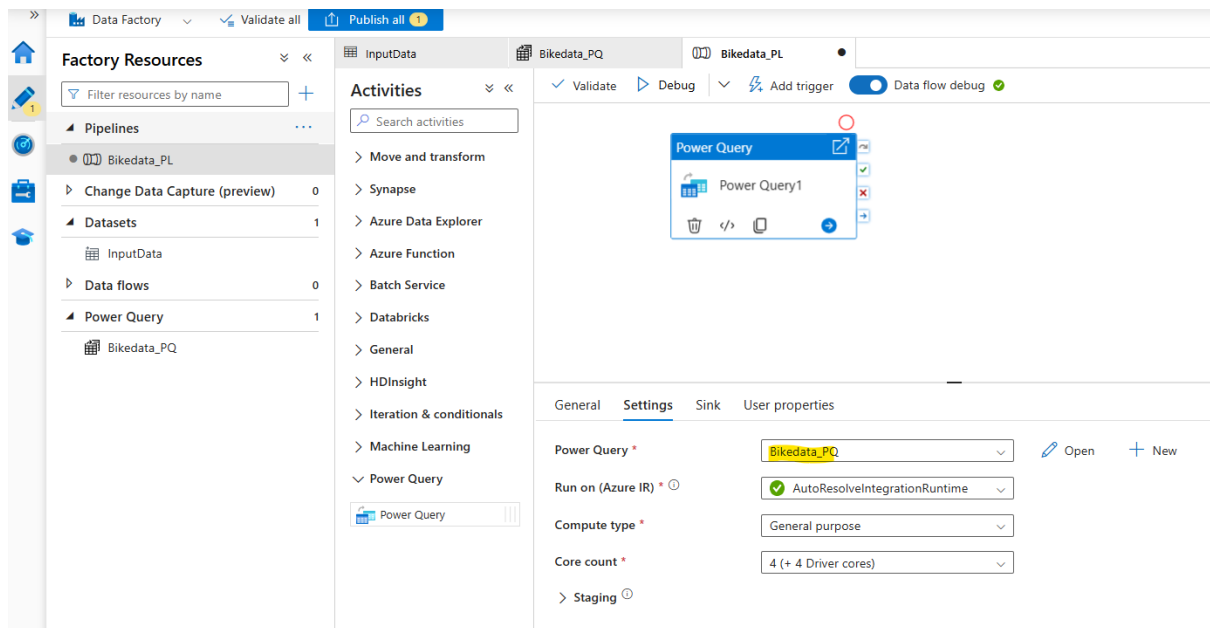
In power query there is no sink so directly we can move to the pipeline

Create Pipeline:

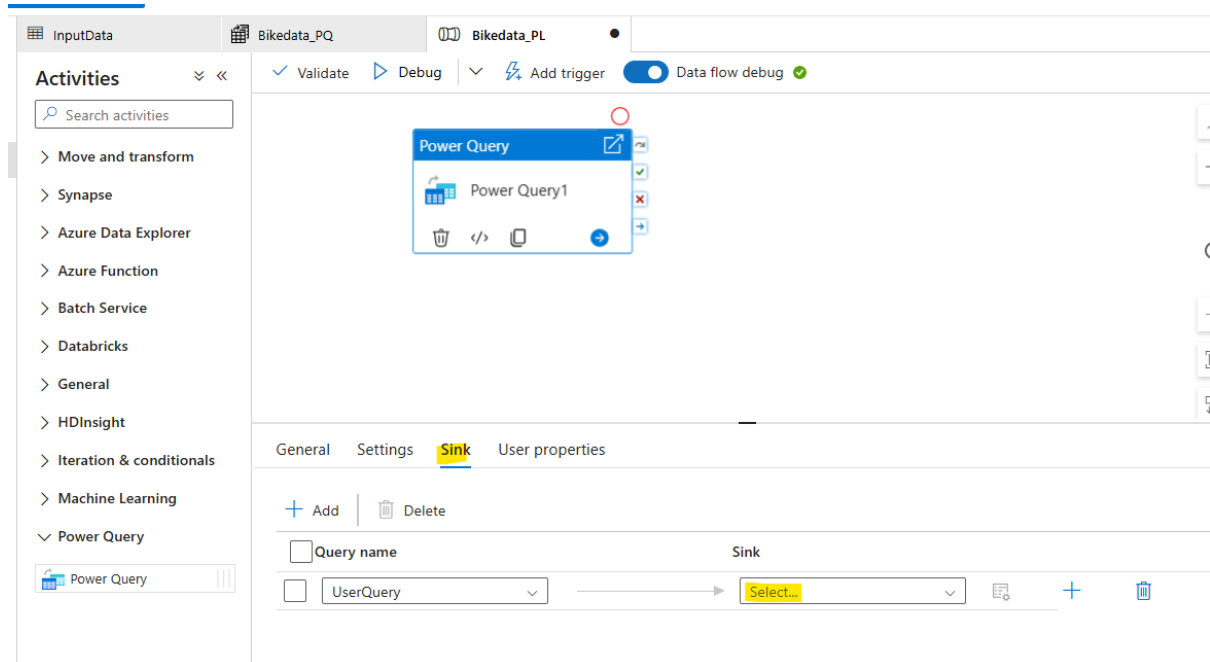


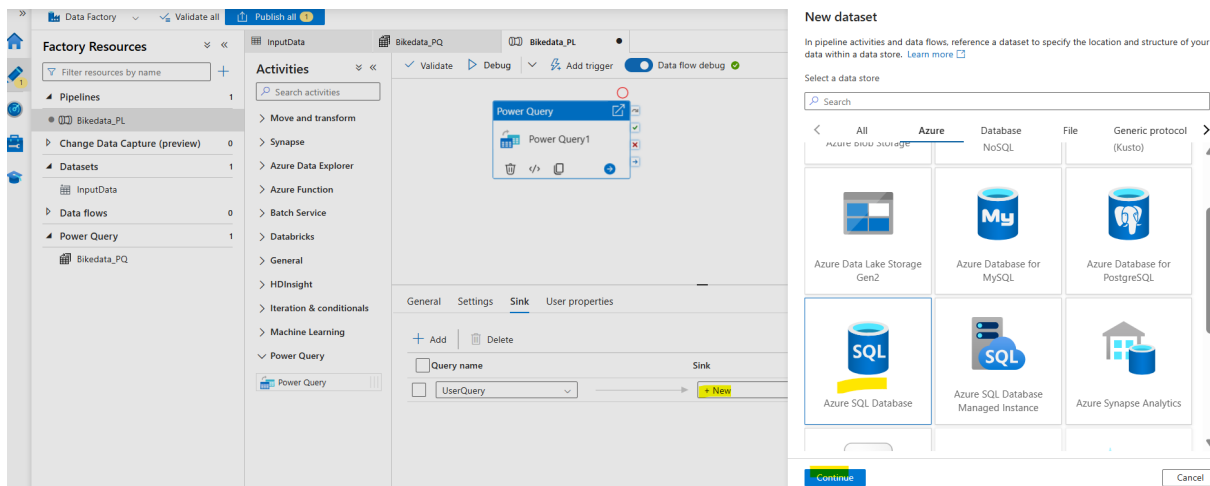
Select the powerquery in the settings





*****Hence we don't have sink in the powerquery in the pipeline---Go to sink---
create new





- Create a new linked service between adf to azure sql db
- Mention all the details like subscription, server name, db name, sqlauthentication(username and password)---test connection

New linked service

Azure SQL Database [Learn more](#)

New linked service

Azure subscription

NK (5bd8f1cd-03ad-48fa-8410-6e58274fc11f)

Server name *

nkserver

Database name *

nkdb

Authentication type *

SQL authentication

User name *

adminnk

Password *

.....

Always encrypted ☐

Encrypt

Mandatory

Trust server certificate ☐

Host name in certificate

Create **Cancel**

Connection successful

Test connection

We don't have any table in azure sql db----need to create a new table---give name

Set properties

Name

Bikedata_cleaned

Linked service *

adf2azsqldb

☐ Select from existing table ☒ New table

Schema and table name

Schema name

Bikedata_TB

OK

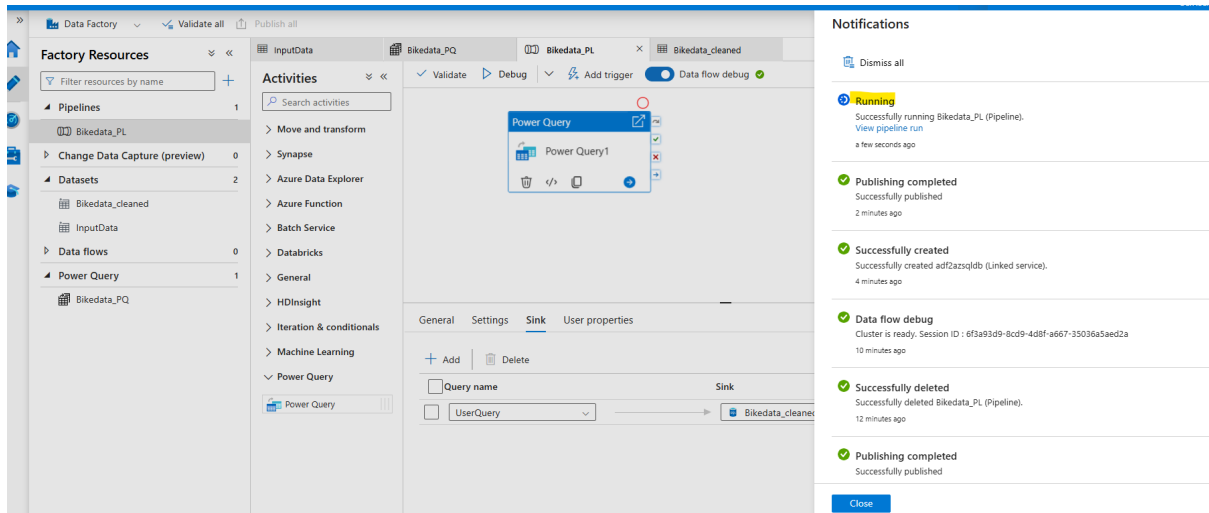
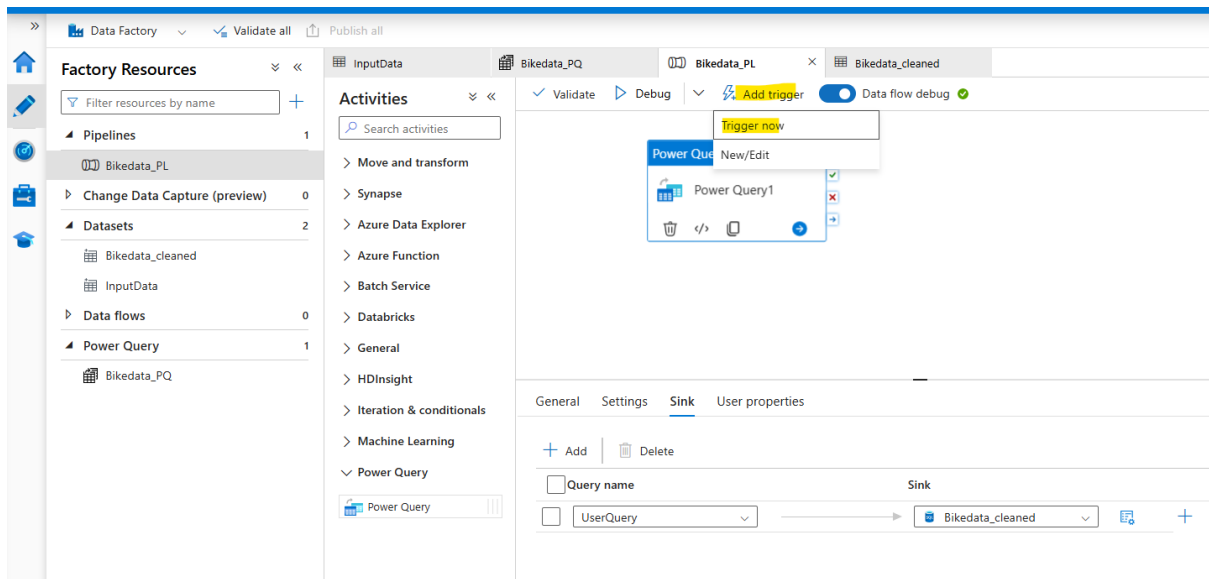
Back

Cancel

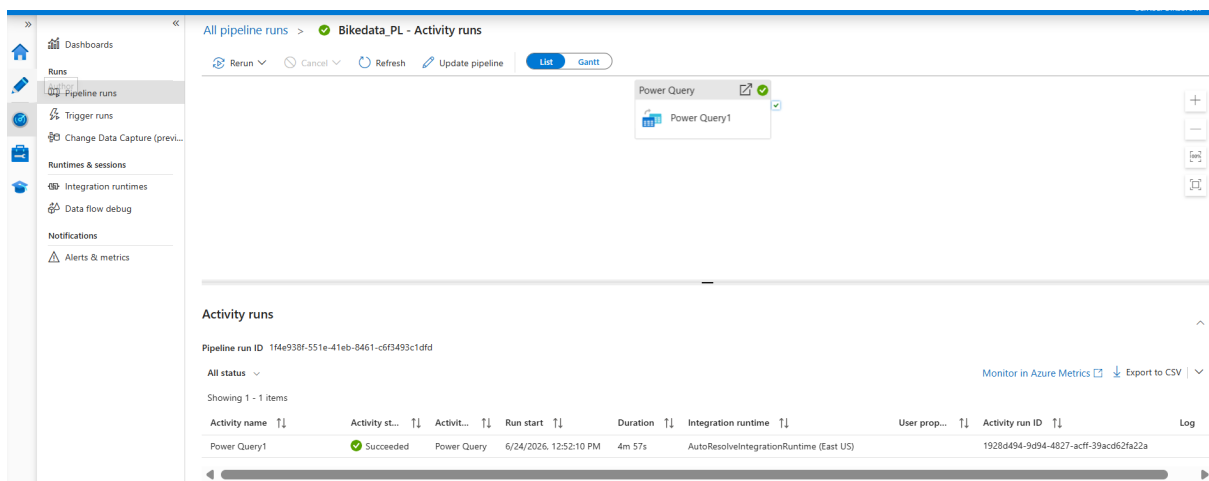
Publish all the changes in the adf---before executing the pipeline

The screenshot displays the Azure Data Factory (ADF) interface. On the left, the 'Factory Resources' pane shows a list of resources including Pipelines, Datasets, and Power Query. The 'Activities' pane in the center lists various activities like Move and transform, Synapse, Azure Data Explorer, etc. The main canvas shows a 'Power Query' activity named 'Power Query1'. Below the canvas, the 'Sink' configuration is visible, showing a dropdown for 'UserQuery' and a dropdown for 'Bikedata_cleaned'. The 'Properties' pane on the right shows the 'General' tab with fields for 'Name' (Bikedata_PL) and 'Description'.

Now add trigger----triggernow—to execute the pipeline



Sucessfully the pipeline has been executed



Check in Azure sql db :

Home > Microsoft SQL Database > newDatabaseNewServer_c09de2497ec74a0bbc0d1_1_Overview > nkdb (nkserver/nkdb)

nkdb (nkserver/nkdb) | Query editor (preview) ☆ ...

SQL database

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Views
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Functions
Queries
SQL query 1

SQL query 1

Run Save as view

1 SELECT TOP (1000) * FROM [dbo].[bikedata_TB]

Messages Results

	ABC	Region	ABC	Country	ABC	Customer	ABC	Business...	ABC	Category	ABC	Model	ABC	SalesDate	12L	OrderQty	12F	Cost
1		North America		U.S.A		Trailblazing S...		Components		Mountain Fra...		HL Mountain ...		24-09-2017		2		1637.4
2		North America		U.S.A		Trailblazing S...		Accessories		Helmets		Sport-100		24-09-2017		4		80.76
3		North America		U.S.A		Extended Bik...		Clothing		Socks		Mountain Bik...		24-09-2017		5		28.5
4		North America		U.S.A		Extended Bik...		Components		Mountain Fra...		HL Mountain ...		24-09-2017		1		722.59
5		North America		U.S.A		Trailblazing S...		Clothing		Caps		Cycling Cap		23-09-2017		4		20.76
6		North America		U.S.A		Extended Bik...		Bikes		Mountain Bikes		Mountain-100		23-09-2017		2		1700

Succeeded (1 sec 794 ms) Columns: 11 Rows: 1000

Successfully the cleaned data is moved to target ---Azure SQL db